TS 101 625 V6.0.0 (1999-01)

Technical Specification

Digital cellular telecommunications system (Phase 2+); High Speed Circuit Switched Data (HSCSD); Stage 1 (GSM 02.34 version 6.0.0 Release 1997)

Reference
DTS/SMG-010234Q6 (f1o03003.PDF)

Keywords
Digital cellular telecommunications system,
Global System for Mobile communications (GSM)

ETSI

Postal address
F-06921 Sophia Antipolis Cedex - FRANCE

Office address
650 Route des Lucioles - Sophia Antipolis
Valbonne - FRANCE
Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16
Siret N° 348 623 562 00017 - NAF 742 C
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° 7803/88

Internet
secretariat@etsi.fr
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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Special Mobile Group (SMG).

This TS defines specifies the Stage 1 description of High Speed Circuit Switched Data (HSCSD) within the digital cellular telecommunications system (Phase 2+).

The contents of this TS is subject to continuing work within SMG and may change following formal SMG approval. Should SMG modify the contents of this TS, it will be re-released with an identifying change of release date and an increase in version number as follows:

Version 6.x.y

where:

6 indicates Release 1997 of GSM Phase 2+

x the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

y the third digit is incremented when editorial only changes have been incorporated in the specification.

1 Scope

This Technical Specification (TS) specifies the Stage 1 description of High Speed Circuit Switched Data (HSCSD). HSCSD is a feature that allows users subscribing to the General Bearer Services to access user rates that can be achieved with one or more TCH/F. HSCSD also defines a flexible use of air interface resources which makes efficient and flexible use of higher user rates feasible.

2 Normative references

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies.
- A non-specific reference to an ETS shall also be taken to refer to later versions published as an EN with the same number.

[1] GSM 01.04: "Digital cellular telecommunications system (Phase 2+); Abbreviations and acronyms".

[2] GSM 02.02: "Digital cellular telecommunications system (Phase 2+); Bearer Services (BS) supported by a GSM Public Land Mobile Network (PLMN)".

[3] GSM 02.04: "Digital cellular telecommunications system (Phase 2+); General on supplementary services".

[4] GSM 02.06: "Digital cellular telecommunications system (Phase 2+); Types of Mobile Stations (MS)".

[5] GSM 05.02: "Digital cellular telecommunications system (Phase 2+); Multiplexing and multiple access on the radio path".

3 Abbreviations and definitions

3.1 Abbreviations

For the purposes of this TS, the following abbreviations apply:

HSCSD	High Speed Circuit Switched Data
MO	Mobile Originated
MT	Mobile Terminated
T	Transparent
NT	Non Transparent
AIUR	Air Interface User Rate
FNUR	Fixed Network User Rate

Additional GSM related abbreviations can be found in GSM 01.04 [1].

3.2 Definitions

For the purposes of this TS, the following definitions apply:

Fixed Network User Rate: The user rate between IWF and the fixed network.

Air Interface User Rate: The user rate between Mobile Termination and IWF. For T services it is the maximum possible AIUR not including padding. For NT services it is the maximum possible AIUR.

maximum possible AIUR: The highest possible AIUR that the multiple TCH/F can provide, e.g. 2 TCH/F using TCH/F9.6 provides a maximum possible AIUR of 19,2 kbit/s.

padding: Fill bits needed to adapt the maximum possible AIUR supported by a given number of TCH/F with a given channel coding to a FNUR that is lower than the maximum possible AIUR.

4 Description

4.1 General

HSCSD is a feature that allows users subscribing to the General Bearer Services to access user rates that can be achieved with one or more TCH/F. HSCSD also defines mechanisms for the use of air interface resources which makes efficient and flexible use of higher user rates feasible.

4.2 Applicability

HSCSD shall provide flexible ways of supporting GSM Phase 2, T and NT data services and new data services at the higher rates possible with one or more TCH/F. Data compression shall be applicable to NT HSCSD. Supplementary Services that are applicable to the General Bearer Services can be used with the HSCSD feature (Reference GSM 02.04, [3]).

4.3 General Bearer Services

The General Bearer Services are defined in GSM 02.02 [2].

The General Bearer Services consist of four Bearer Services, and they are as follows:

- asynchronous;
- synchronous;
- dedicated PAD access;
- dedicated packet access.

4.4 Parameters to be indicated and negotiated

4.4.1 Call set-up

The parameters to be indicated, and negotiated, if applicable, during the call set-up shall include:

- FNUR;
- Channel coding(s) acceptable (for the call);
- Maximum number of TCH/F (that the mobile user can accept);

- Wanted AIUR (desired rate that the mobile user wants the network to allocate). The wanted AIUR is applicable to NT services only;
- Indication if the user initiated modification is required, and if so, the network resource needs.

The channel coding(s) acceptable shall be indicated by the mobile at call set-up and is not negotiable. It indicates the channel coding(s) that may be chosen by the network for the call.

The maximum number of TCH/F shall be indicated at call set-up. It enables the mobile user to limit the number of TCH/F used and thus to control an essential parameter for charging. It sets the upper limit of number of TCH/F that the network may allocate to the mobile.

The wanted AIUR (applicable to NT services only) indicates the AIUR that the mobile user wants and which the network shall try to reach but which it is not allowed to exceed. The exception where the network is allowed to exceed the wanted AIUR is when the network can achieve the AIUR with a lower number of TCH/F, e.g. wanted AIUR indicated by the mobile is 14,4 kbit/s, channel codings acceptable are both TCH/F9.6 and TCH/F4.8 and maximum number of TCH/F are 3, then the network shall choose 2 x 9,6 over 3 x 4,8 if a channel coding of TCH/F9.6 is available on two TCH/F.

If the user wishes to make use of the user initiated modification procedure, this shall be indicated at the call set-up.

It shall be possible to reserve a FNUR that is considerably higher than the AIUR.

4.4.1.1 T services

The ME or the network may propose to modify the FNUR. The calling entity may accept or release the call. Autobauding is not allowed.

The AIUR is always equal to the FNUR.

Fax Group 3 can make use of HSCSD.

The channel coding selected must be one of the channel coding(s) indicated in the channel coding(s) acceptable parameter, the number of TCH/F selected shall not exceed the maximum number of TCH/F parameter, and the combination of the two shall result in an AIUR that is equal to the FNUR.

4.4.1.2 NT services

The ME or the network may propose to modify the FNUR. The calling entity may accept or release the call. Autobauding is allowed.

The wanted AIUR is indicated by the mobile at call set-up.

4.4.2 Network initiated modification

4.4.2.1 T services

For transparent calls the radio resource parameters must remain within limits that allow the transparent call to maintain it is characteristics of fixed end to end throughput and delay. This means that the channel coding and the number of TCH/F used may change during the call as long as a channel coding indicated in the channel coding(s) acceptable parameter is used, the maximum number of TCH/F is not exceeded and the AIUR is kept constant.

4.4.2.2 NT services

For NT calls the network may modify the number of TCH/F and the channel coding used and thus also the AIUR during the call as long as the maximum number of TCH/F, and the channel coding(s) acceptable are all respected. The network shall try to reach the wanted AIUR indicated, as long as the resource situation allows it.

4.4.3 User initiated modification

The in-call modification feature shall be supported in case of alternate services.

4.4.3.1 T services

The user initiated modification is not applicable to T services.

4.4.3.2 NT services

The user may indicate a change of the parameters maximum number of TCH/F and wanted AIUR only.

The user initiated modification is only applicable to the data phase of alternate services.

4.5 Air Interface Resource Allocation

4.5.1 Minimum Air Interface Resource Allocation and Allocation Increment

Air interface resources shall be allocated to HSCSD calls at TCH/F increments. The minimum air interface resource allocation for HSCSD calls shall be one TCH/F.

4.5.2 Flexible Air Interface Resource Allocation

Flexible air interface resource allocation enables:

> The network to allocate dynamically resources related to the air interface usage according to the network operator's strategy, within the limits negotiated at the call set-up or during a user initiated modification.

4.6 Symmetric and asymmetric connections

4.6.1 Symmetric connections

For symmetric connections, air interface resources are allocated symmetrically.

4.6.2 Asymmetric connections

The provision of the asymmetric air interface connections allows simple ME (of Type 1, Reference GSM 05.02, [5]) to receive at higher AIUR than otherwise would be possible with a symmetric connection.

Asymmetric air interface connections that are a subset of the symmetric HSCSD, and support different user rates at uplink and downlink, shall be provided.

Asymmetric air interface connections are only applicable to the downlink-biased asymmetry, i.e., where the ME is receiving at a higher rate than it is transmitting.

Asymmetric air interface connections shall only be applicable to NT HSCSD.

4.6.3 Network choice of connection symmetry

When the network has a choice of allocating either a symmetric or an asymmetric air interface connection it shall proceed as follows:

- In the case where the wanted AIUR is smaller than or equal to the AIUR supported symmetrically by the ME, or asymmetric air interface connection is not supported by the network, then a symmetric air interface connection is established.

- In the case where the wanted AIUR exceeds the AIUR supported symmetrically by the ME, the network shall assign the maximum AIUR supported by the ME in the down-link, upper-bounded by the maximum number of TCH/F indicated by the mobile user, and the number of TCH/F supported by the network, maintaining the minimum of one TCH/F.

4.7 Mobile Station/Mobile Equipment requirements

The general MS capabilities is applicable to HSCSD (Reference GSM 02.60 [4]).

A ME that supports HSCSD shall support one or more of the channel types TCH/F at or above 4,8 kbit/s.

Given that the ME supports a certain channel coding, the network shall be able to assume that the ME can support this channel coding in any multislot configuration allowed by its multislot class (Reference GSM 05.02, [5]).

4.8 Mobility Management

HSCSD shall be provided for within the existing supported Mobility Management.

4.9 Roaming

Roaming shall be possible.

4.10 Quality of Service

4.10.1 Bit error rate

4.10.1.1 TCH/F Bit Error Rate

The bit error rate performance of each TCH/F of a multiple TCH/F HSCSD call shall be the same as that of a single TCH/F data call for the same channel coding.

4.10.1.2 Overall HSCSD Bit Error Rate

Any increase in the bit error rate caused by the splitting and combining of multiple TCH/F shall be kept to a minimum.

4.10.2 Radio resource management

HSCSD shall be provided for within the existing Radio Resource management control functions, including the relevant handover types. The handover shall be simultaneous for all air interface timeslots making up an HSCSD call.

HSCSD shall be provided with full mobility.

4.10.3 Call set-up delay

Any increase in call set-up delay of the HSCSD calls compared to Phase 2 data services shall be kept to a minimum.

4.10.4 End-to-end delay

Any increase in end-to-end delay of the HSCSD calls compared to Phase 2 data services shall be kept to a minimum.

4.10.5 Throughput

The throughput for HSCSD T calls shall remain constant for the duration of the call except for the interruption of transmission at handover.

4.10.6 Network planning and interference levels

The impact of HSCSD on network planning and interference levels shall be kept to a minimum.

4.11 Mapping of FNUR to TCH/F for T services

For some FNUR padding is required to adapt the maximum possible AIUR to the FNUR.

4.12 Mapping of AIUR to TCH/F for NT services

The following table indicates the mapping of AIUR to the number of TCH/F for NT services.

Table 1/GSM 02.34: Mapping of AIUR to TCH/F for NT services

AIUR	TCH/F4.8	TCH/F9.6	TCH/F14.4
4,8 kbit/s	1	N/A	N/A
9,6 kbit/s	2	1	N/A
14,4 kbit/s	3	N/A	1
19,2 kbit/s	4	2	N/A
28,8 kbit/s	N/A	3	2
38,4 kbit/s	N/A	4	N/A
43.2 kbit/s	N/A	N/A	3
57.6 kbit/s	N/A	N/A	4

N/A Not applicable.

4.13 HSCSD MMI aspects

It shall be possible to configure the parameters negotiated and indicated at HSCSD call set-up and during user initiated modification.

4.14 HSCSD Subscription aspects

HSCSD is not subscribed to. Users wanting to use HSCSD must subscribe to the General Bearer Services.

4.15 HSCSD charging aspects

To allow accurate charging of HSCSD calls, a record of start and stop timestamps versus the number of TCH/F, for each change in TCH/F allocation, shall be provided.

NOTE: For MO HSCSD calls, the A party is liable for the use of all the TCH/F in her PLMN.

For MT HSCSD calls, the B party may have to pay for one or more of the TCH/F in her PLMN.

In case the originating or terminating subscriber is a fixed network subscriber, there should be no additional charge (in respect of the changing allocation of TCH/F) for the originating or terminating fixed network subscriber.

4.16 O and M aspects

For HSCSD call, there are several new and modified parameters compared to GSM Phase 2 data calls.

For HSCSD, the air interface resource allocation may change several times during the call (network initiated and user initiated changes). It shall be possible to limit the frequency of changes per call, separately for network initiated and user initiated changes.

HSCSD calls can occupy multiple TCH/F. It shall dynamically be possible to limit the maximum number of TCH/F allowed for HSCSD calls per cell.

4.17 HSCSD security aspects

Authentication and ciphering are applicable to HSCSD calls with no or minimal reduction in the security of the air interface.

5 Exceptional procedures or unsuccessful outcome

When a HSCSD call is offered to an entity (either ME or network) unable to support the offered HSCSD call, it shall be possible, within the limitations of the service, to revert the call:

- to an HSCSD call the entity can support; or
- to a GSM Phase 2 data service.

6 Interworking requirements

6.1 Service interworking

6.1.1 Interworking with supplementary services

Supplementary Services that are applicable to the General Bearer Services can be used with the HSCSD feature (Reference GSM 02.04, [3]).

6.1.2 Interworking with other services

The simultaneous use of HSCSD and SMS MO/PP, and HSCSD and SMS MT/PP services shall be possible.

6.2 Network interworking

Interworking should be possible to all networks which are covered by GSM Phase 2 bearer services:

- PSTN;
- ISDN;
- PSPDN;
- CSPDN.

Annex A (informative): Change history

Change history					
SMG No.	Tdoc. No.	CR. No.	Section affected	New version	Subject/Comments
SMG#21	168/97	A001 A002 A003 A004r1 A005 A006	Various 4.5.1,4.15 4.5.2,4.16 4.10.6,4.15,6.2 4.6.2 4.7	5.1.0	HSCSD calls, MEs, abbreviations. Resource "allocation", "reservation" and "use" terminology. Description of flexible air interface resource allocation. Normative and informative text. Rates with asymmetric connections. Support of Multi band MS.
SMG#22	T1P1.5/97-011R2	A007	4.12 table 1	5.2.0	Adding the TCH/F14.4 Column, adding the 43.2 Kbps and 57.6 Kbps in the AIUR Column
SMG#27				6.0.0	Upgrade to version 6.0.0 as Release 1997

History

Document history		
V6.0.0	January 1999	Publication

ISBN 2-7437-2770-5
Dépôt légal : Janvier 1999